

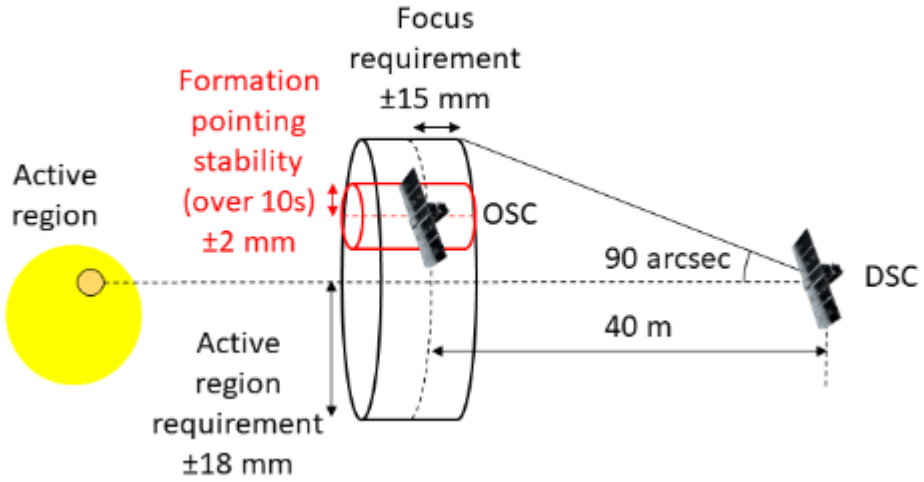
AA 273 FINAL PROJECT

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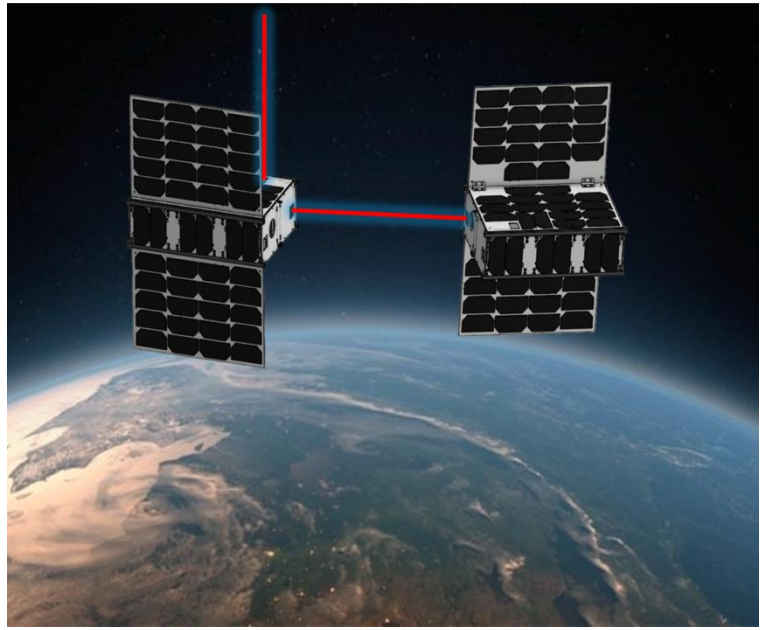
Relative State Estimation using LED Beacons

Introduction

- Satellite formations impose stringent relative pose requirements for mission objectives and safety
  - Relative position requirements on the order of 1mm over 100-m baselines
- **Imperative to use a low-SWaPC method of inter-satellite ranging for real-time pose estimation**
- Useful for swarms with science objectives that need tight positioning (STARI, VISORS) or for RPOD operations



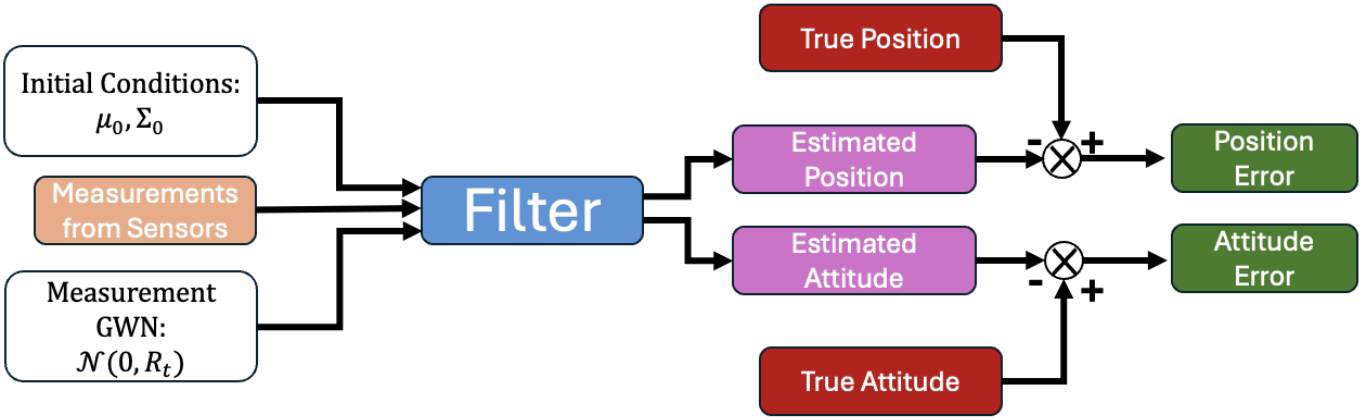
The VISORS Mission, requiring 90 arcseconds of pointing precision at 40 meters [1]

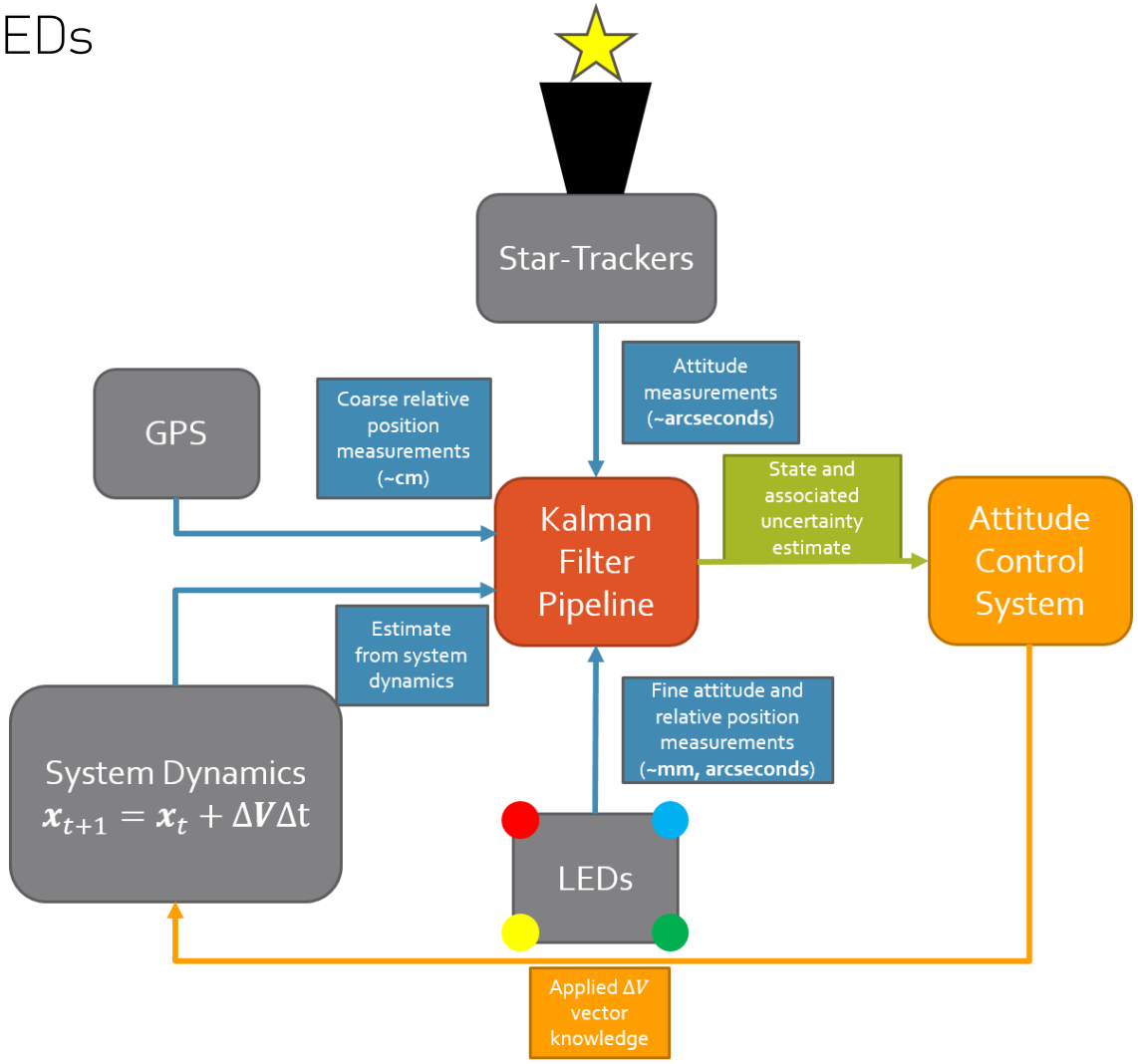


The STARI Mission, aiming to reflect starlight onto a 1-inch square target at a 100-meter distance [2]

Navigation Pipeline

- Sensor suite for position: **Dynamics**, GPS, **LEDs**
- Sensor suite for attitude: Dynamics, Star trackers, LEDs



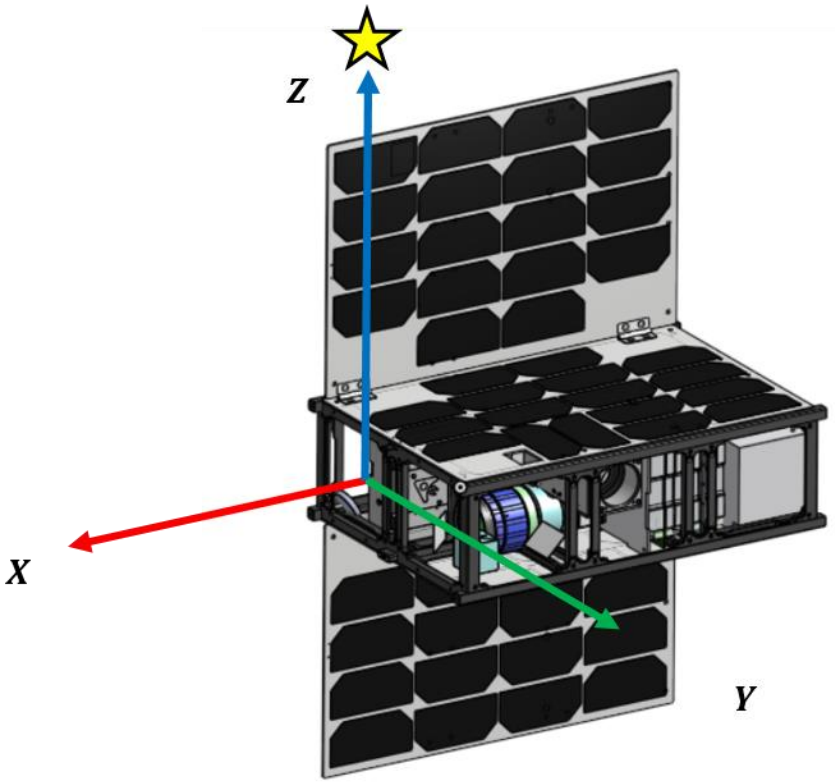


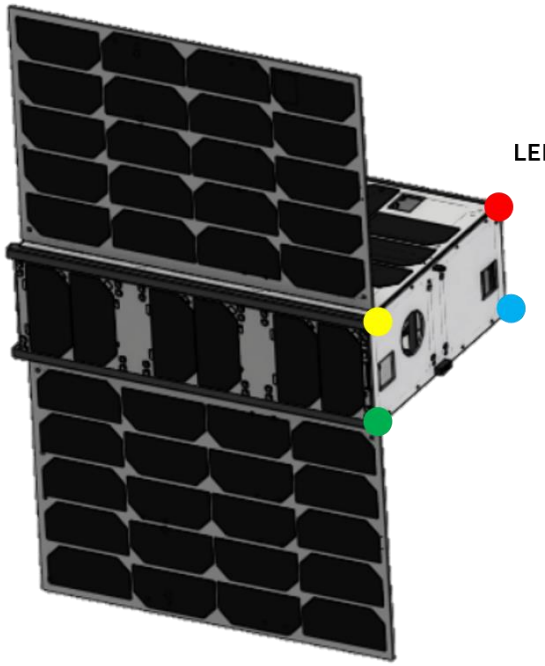
Coordinate System and Terminology

- Distributed Space System – **Chief and Deputy**
- **State vector defined** by relative position and attitude:

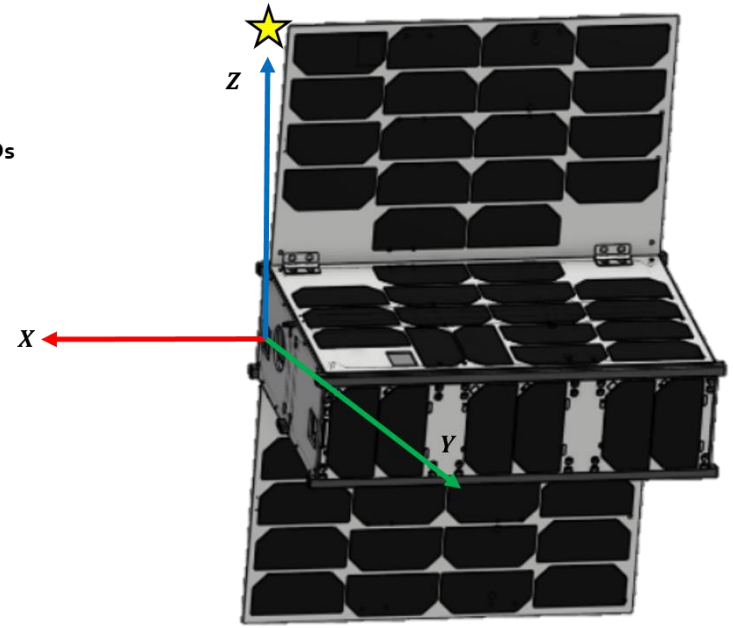
$$\mathbf{x} = \begin{bmatrix} X & Y & Z & \theta & \phi & \psi \end{bmatrix}^T$$

- **Not considering attitude**, would need to represent using quaternions to avoid singularities, and not possible to use existing KFs to maintain unit norm constraint
- **Ignoring perturbations** to simplify dynamics
- **Ignoring control** – purely passive orbits, ignore attitude control





DEPUTY



CHIEF

Dynamics Model + Filter Choice

- Several ways to propagate the dynamics of the satellite:
  - HCW – Linearized circular orbit, propagate relative state
  - YA – Linearized eccentric orbit, propagate relative state
  - ROE STM – Nonlinear eccentric orbit, propagate relative orbital elements
- Chose a **standard KF** over other varieties because our equations are linear and don't require a Jacobian

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{x}_t \qquad \mathbf{B} = \mathbf{0}_{6x3}$$

$$\mathbf{x}_t = \begin{bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}$$

Kalman Filter Predict Step:

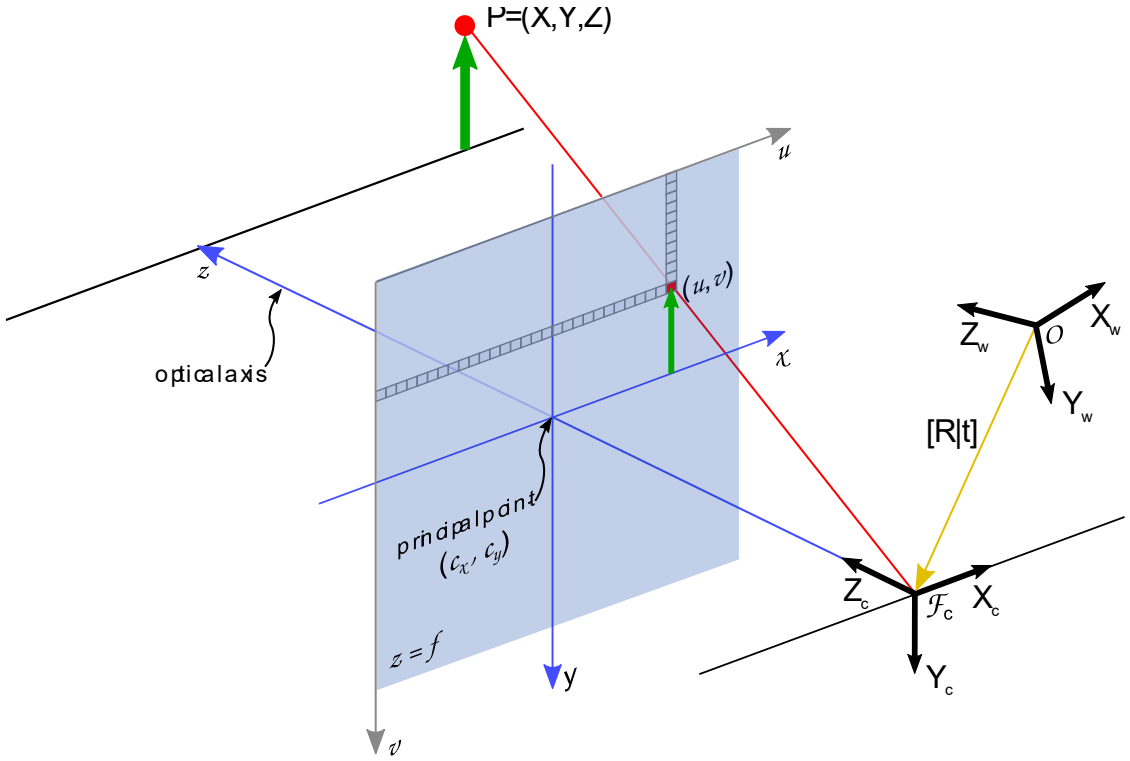
$$\text{mean:} \quad \mu_{k|k-1} = A_{k-1}\mu_{k-1} + B_{k-1}u_{k-1}$$

$$\text{cov:} \quad \Sigma_{k|k-1} = A_{k-1}\Sigma_{k-1}A_{k-1}^T + Q_{k-1}$$

[4][5][6][7]

Camera model

- Using a **pinhole camera model**
- Knowledge of:
  - LED positions on Deputy
  - Detected LED positions in pixels
  - Camera intrinsics matrix K
- Define p^*_x and p^*_y as **pixel pitch**, in m/pixel
- Define f as **focal length**, in meters
  - Define $f_{x,y} = f/p^*_{x,y}$ as focal length in pixels
- Define c_x and c_y as **image center**, in pixels
- Skew value s = 0
- Obtained real parameters for an IDS UI-3060CP camera



$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

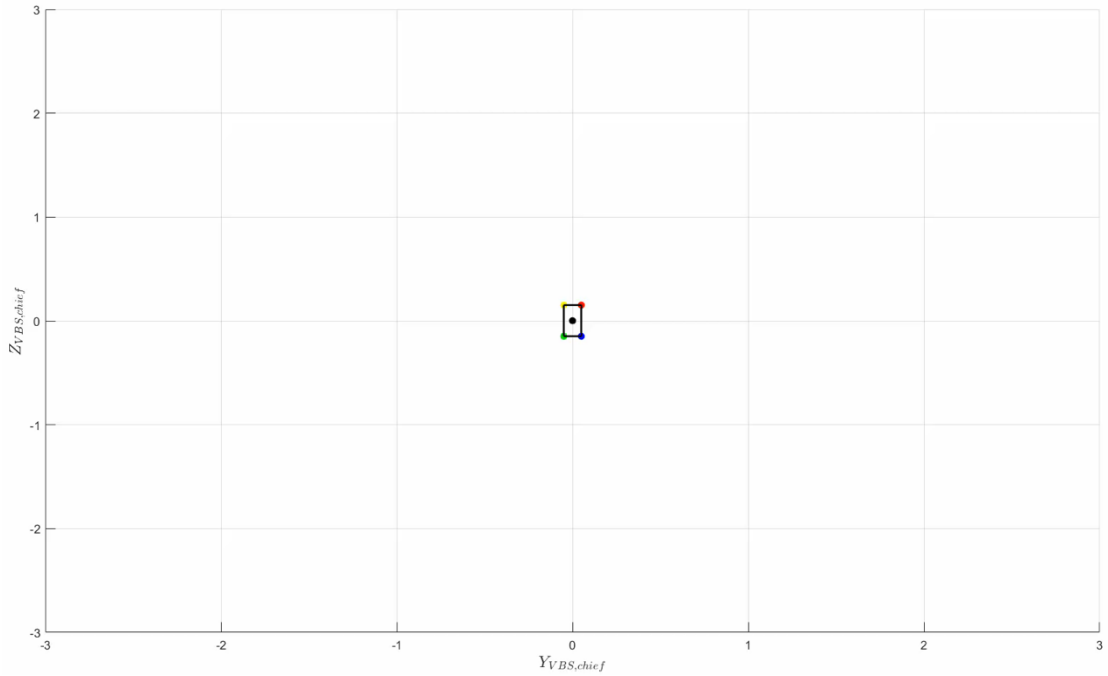
Pinhole camera model:

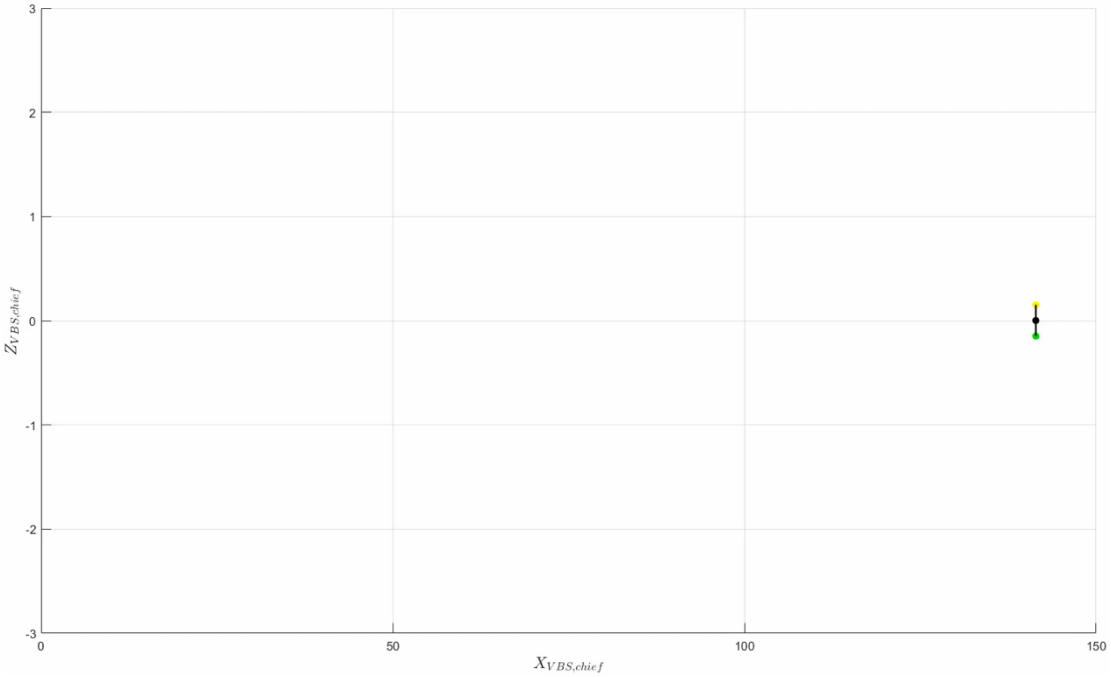
$$p^h = p_z \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

homogeneous coordinates

Measurement Model

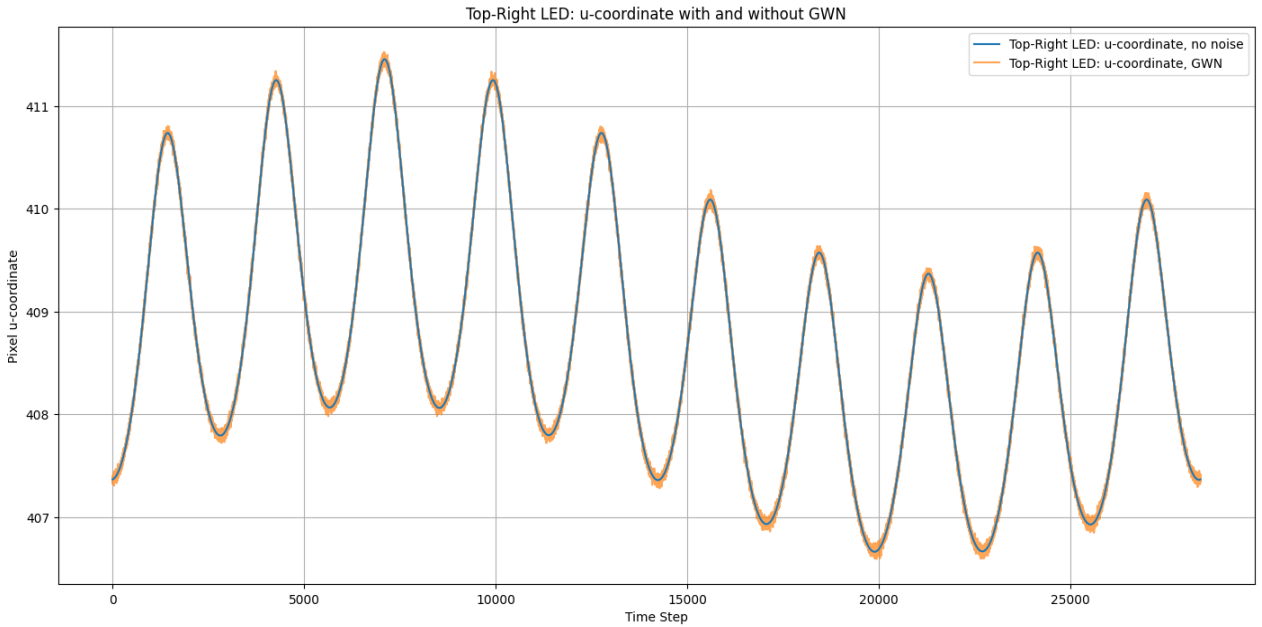
- Using **STARI** simulation datasets
- Programmed in as a **lookup table of relative position**, indexed by time step
  - Simulation data -> Pixel values using inverse of pinhole model
  - Pixel values -> Relative position data using pinhole model
- Central differencing -> Relative velocities
- **Identity Jacobian of measurement model**





Noise Modeling

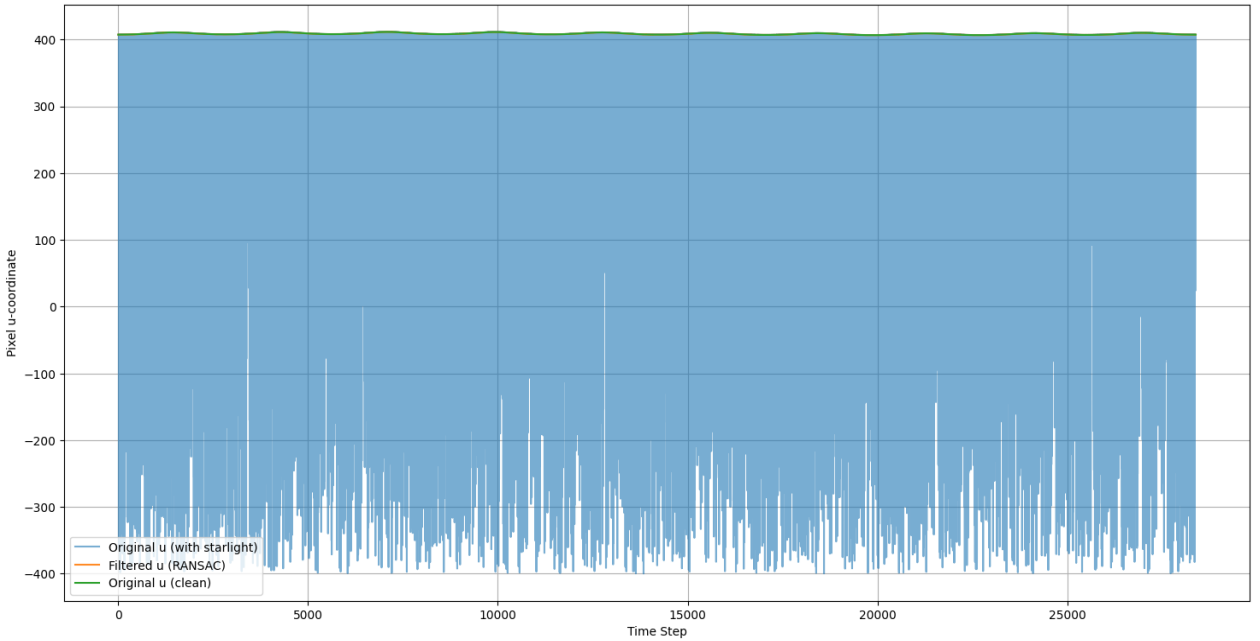
- Worked with 3 distinct data sets:
  - Clean: No injected noise
  - GWN: **Gaussian white noise** injected into pixel measurements; characteristic of real CCDs
    - Covariance = 0.03 (characteristic of typical CCD sensor noise) [3]
  - "Dirty": GWN + **starlight noise**, modeled as a random Poisson process such that a star is mistakenly thought of as an LED

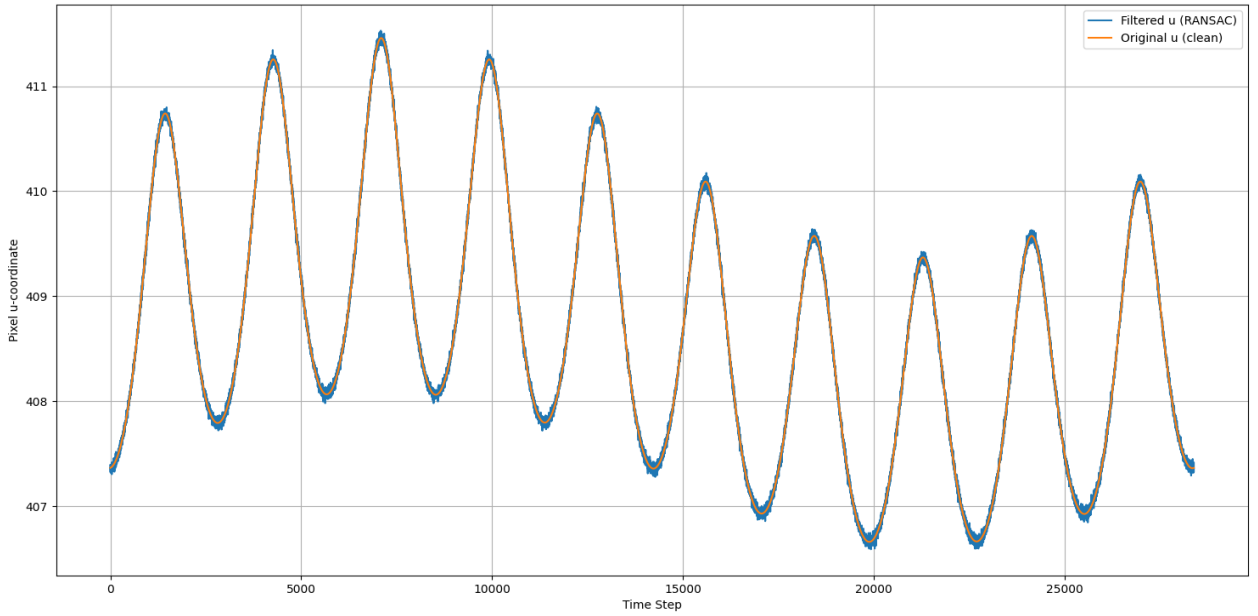


- Note: In real applications, starlight noise is actually helpful and can be referenced against star catalogs to provide an additional sensor measurement!

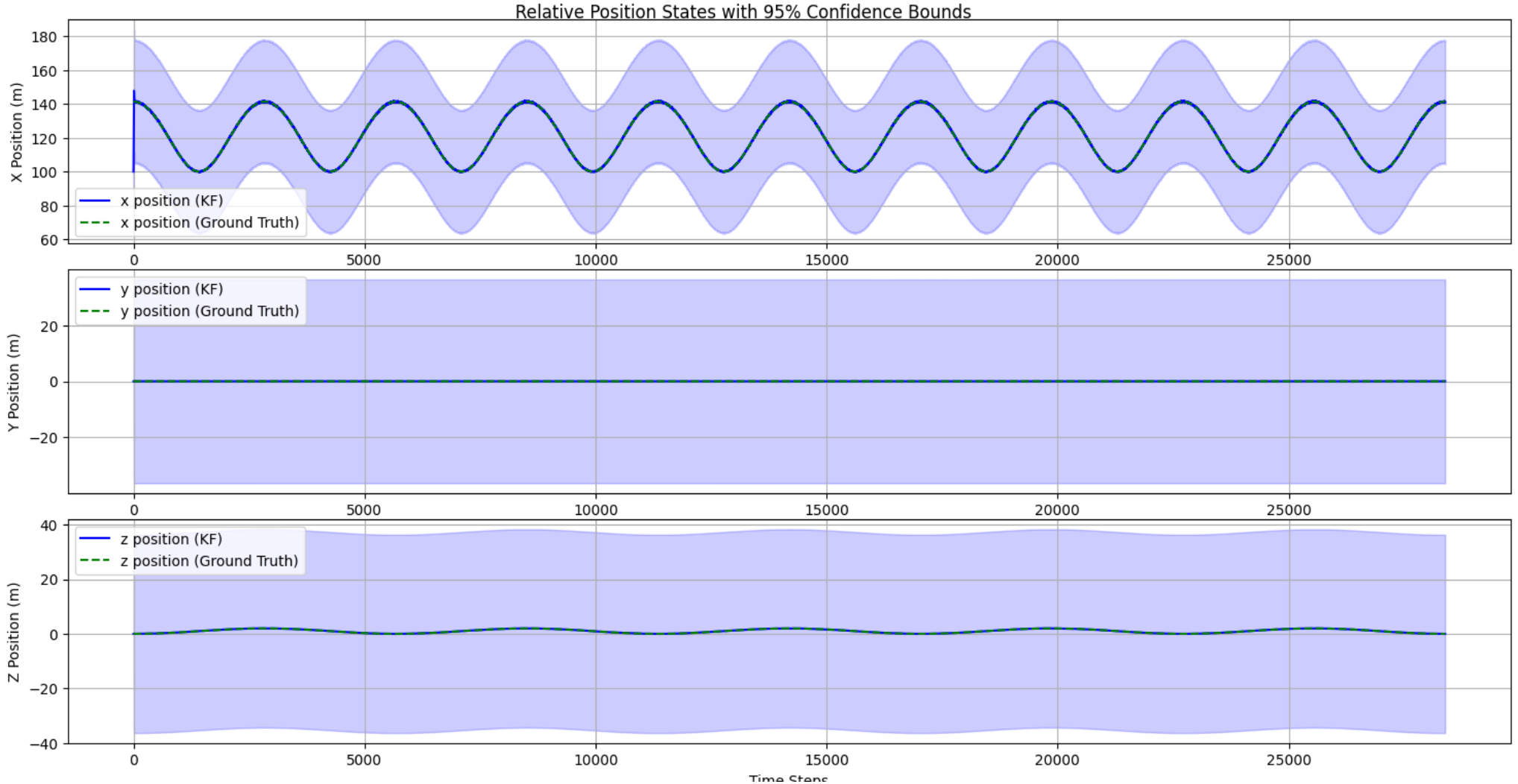
Outlier Rejection

- For "dirty" dataset: Used a form of **online outlier rejection** using **linear regression** for **iteratively removing noise from pixels at each time step**
  - Fits a linear model to all past data
  - Computes **residual at every time step and compares to a fixed threshold value**
    - Points below the threshold are accepted,
    - Points above the threshold are rejected and set to the last accepted value
- Initially considered RANSAC, but this was not suited to online filtering



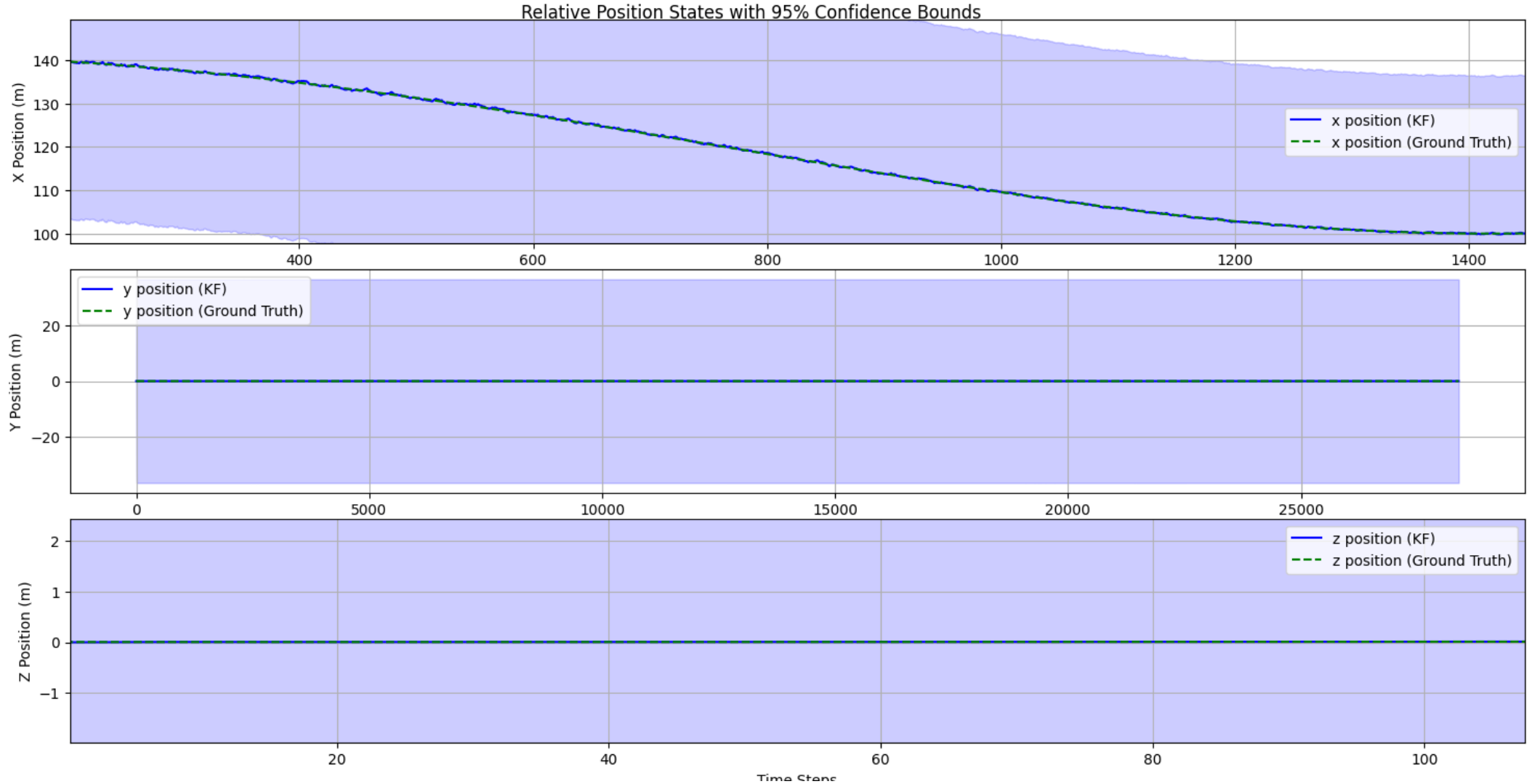


Results – GWN



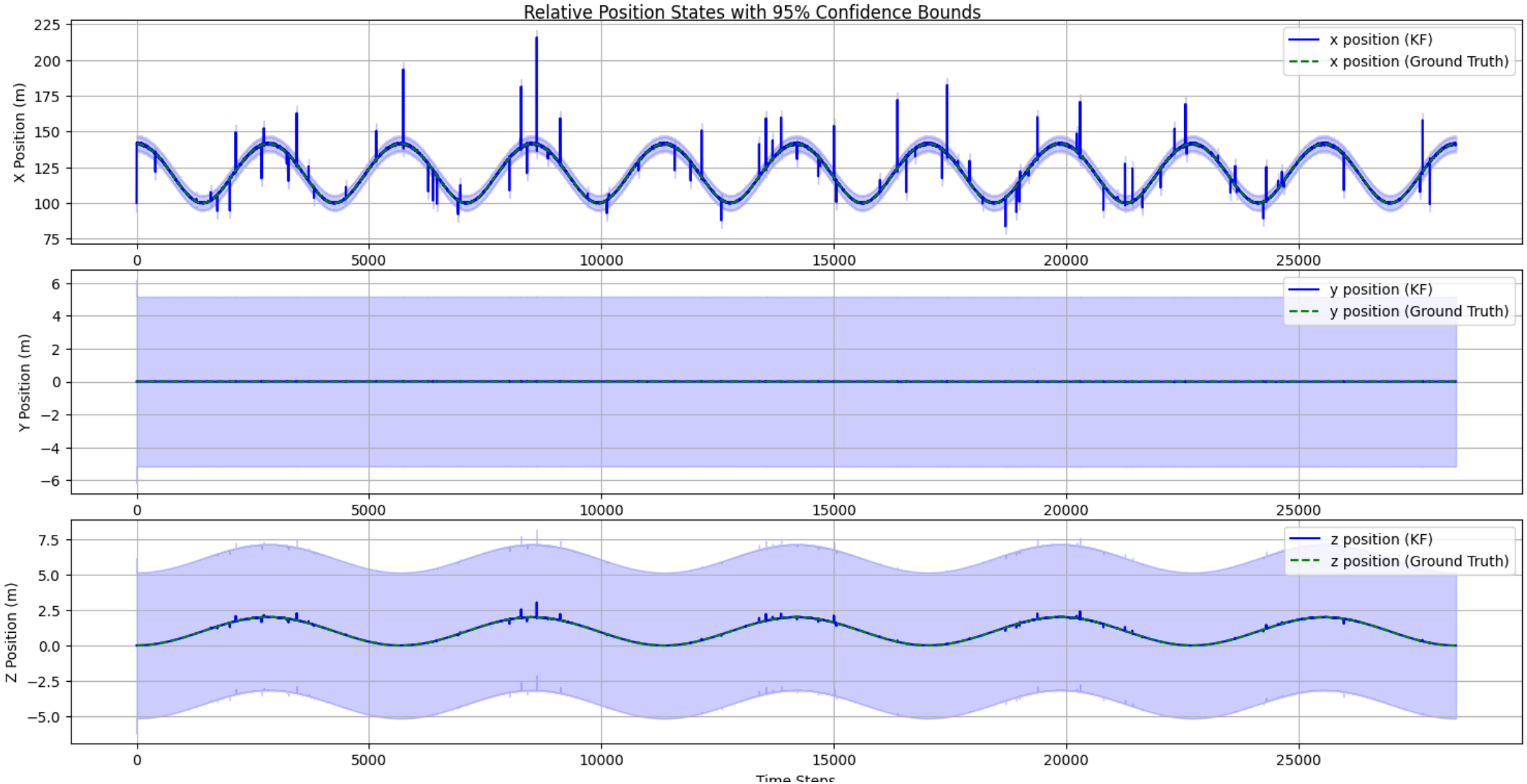
- Q=10, R=1000

Results – GWN



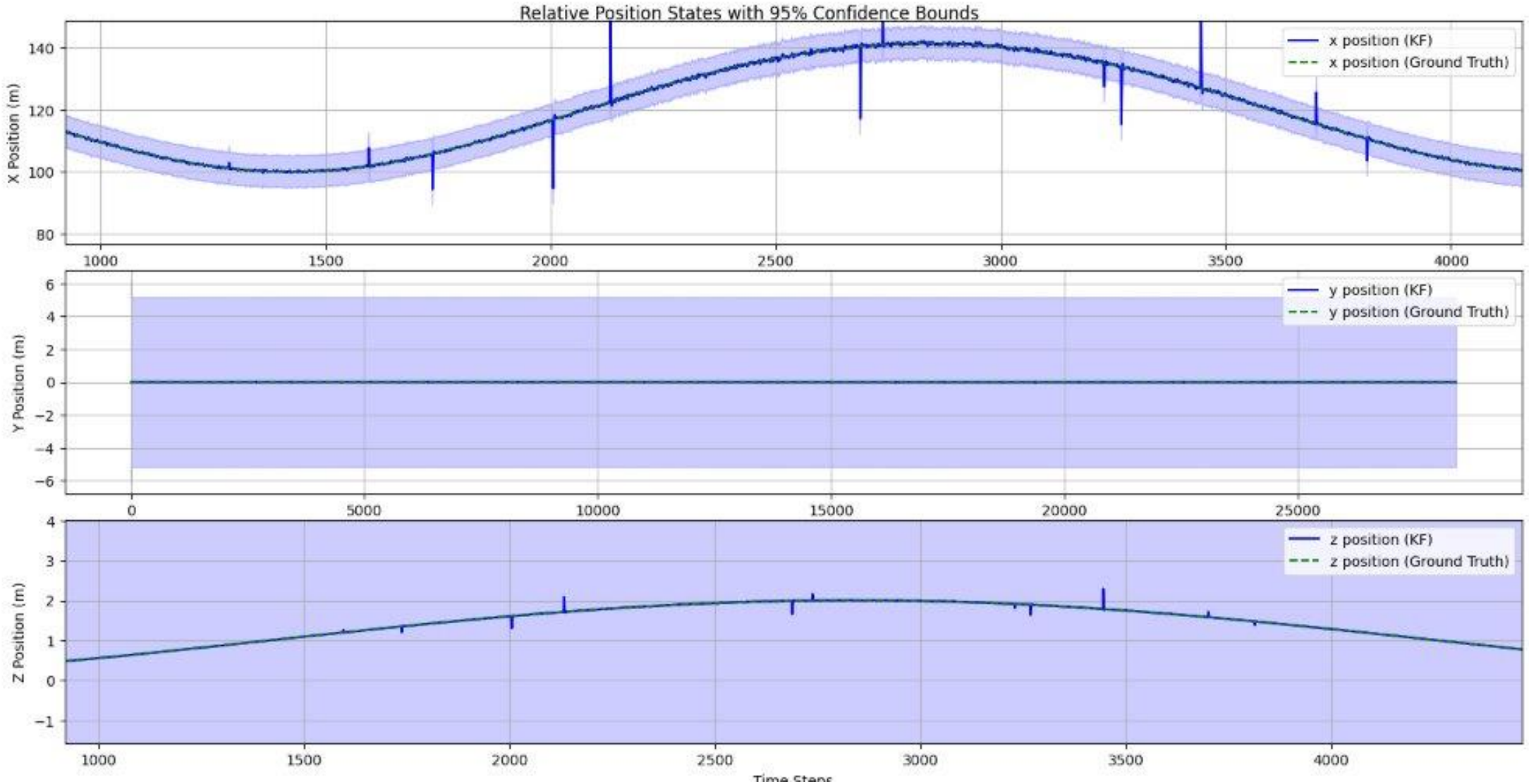
- Errors on the order of <1m in X and <1mm in Z

Results - "Dirty"



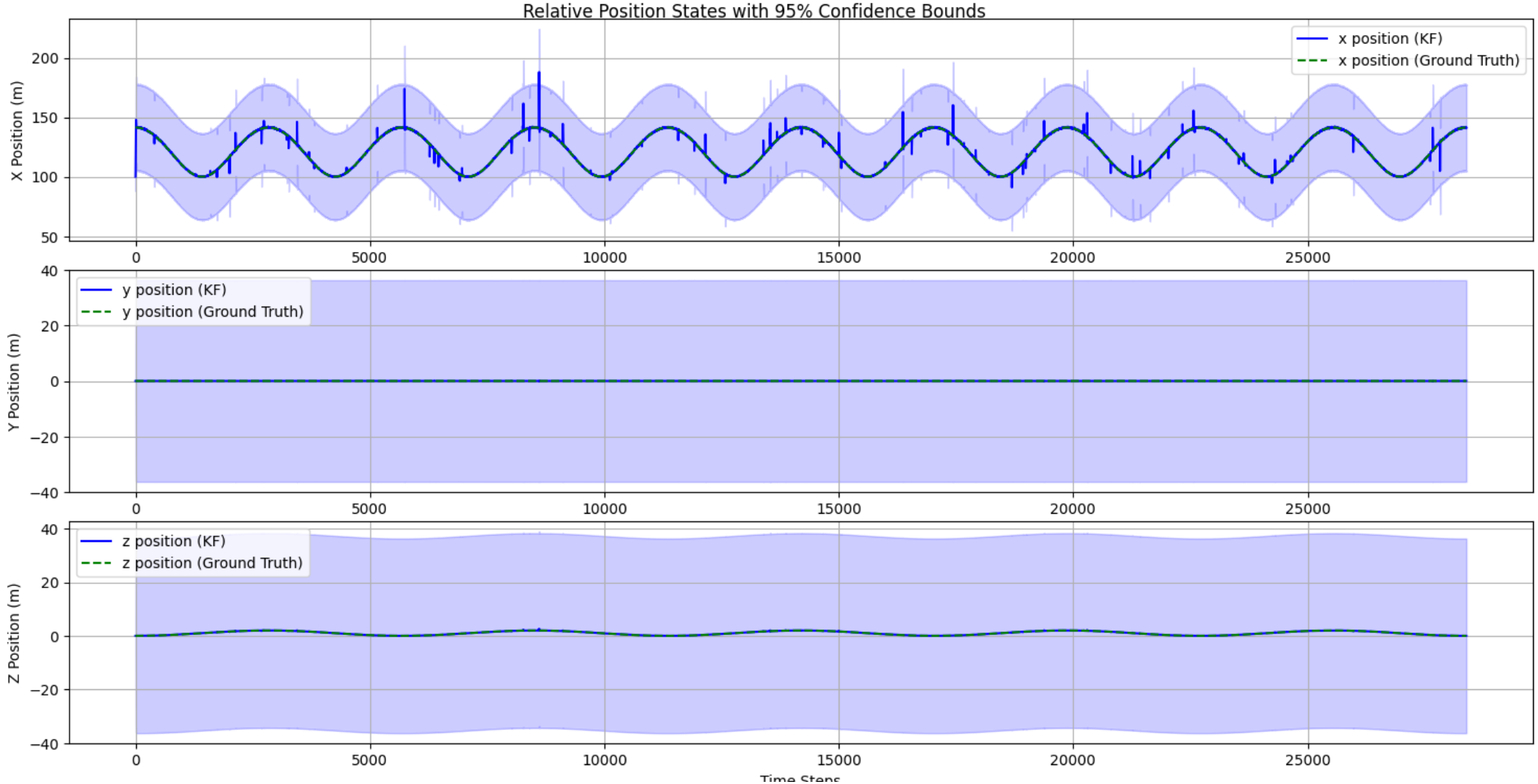
- Intermittent spikes still observed
- Spikes in X => Corresponding spikes in Z

Results - "Dirty"



- Zoomed in view to show remarkable tracking most of the time
- Q, R = 10

Results – "Dirty"



- Increased R to 1000 to understand effect on estimation – spikes decrease in size, CI gets larger
- Errors on the order of 1m in X and 1mm in Z, but with intermittent spikes

Future Work

- More **robust outlier rejection method** for starlight filtering
- Incorporation of **attitude dynamics** for **full pose estimation** using camera homography
- **Sensor fusion** with simulated star-tracker and GPS data
- Leveraging **stars in LED camera FOV** as an **added data stream** for state estimate refinement
- Higher-fidelity orbital dynamics model – use YA or FODE for a more exact solution
- More realistic estimates for Q and R
- Methodology for distinguishing between LEDs (color vs flickering frequencies)

References

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- [2] J. D. Monnier et al., "STARI: starlight acquisition and reflection toward interferometry," Space Telescopes and Instrumentation 2020: Optical, Infrared, and Millimeter Wave, p. 112, Jun. 2024, doi: 10.1117/12.3020785.
- [3] M. V. Konnik and J. S. Welsh, "High-level numerical simulations of noise in CCD and CMOS photosensors: review and tutorial," arXiv (Cornell University), Jan. 2014, doi: 10.48550/arxiv.1412.4031.
- [4] S. D'Amico, Autonomous Formation Flying in Low Earth Orbit, Ph.D.dissertation, Technische Universiteit Delft, Delft, The Netherlands, Mar.2010. [Online]. Available: https://doi.org/10.4233/uuid:df5b3f2e-1d6d-4a4a-b7d7-6f4b6f9c1e0d
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- [6] K. Yamanaka and F. Ankersen, "New State Transition Matrix for Relative Motion on an Arbitrary Elliptical Orbit," Journal of Guidance, Control, and Dynamics, vol. 25, no. 1, pp. 60–66, Jan.–Feb. 2002. doi: 10.2514/2.4875.
- [7] M. B. Willis, Analytical Theory of Satellite Relative Motion with Applications to Autonomous Navigation and Control, Ph.D. dissertation, Department of Aeronautics and Astronautics, Stanford University, Stanford, CA, Mar. 2023.